Project Description

Course: Database Systems
Fall 2025–2026

Overview
The course project aims to apply the concepts learned throughout the semester by designing and implementing a complete relational database system for a realistic scenario of your choice (for example, university registration, library system, clinic management, internships, etc.). Each group will design the database, implement it using SQL, and execute a set of diverse queries demonstrating the database's capabilities.

1. Database Design

a. Conceptual Design

- Build a complete Entity-Relationship (ER) or Enhanced ER (EER) diagram to model your chosen topic.
- Clearly define entities, attributes, primary and foreign keys, relationships, cardinalities, and participation constraints.
- Add relationship attributes, specializations/generalizations, and weak entities.

b. Logical Design

- Map your ER/EER diagram into a complete relational schema.
- Clearly specify table names, attributes and data types, primary and foreign keys, constraints (UNIQUE, DEFAULT, NOT NULL), and referential integrity (ON DELETE / ON UPDATE actions).

2. Query Execution

a. Questions and Queries

Prepare a list of 15 questions that can be answered using your database.

b. SQL Implementation

Translate each question into a corresponding SQL query and execute it. Include the query, the output (screenshot or result), and a short explanation of what each query does.

c. Query Diversity

Your 15 queries must cover the following topics:

- Joins (INNER, LEFT, RIGHT)
- Aggregate functions, ORDER BY, GROUP BY, HAVING
- Nested queries
- EXISTS / NOT EXISTS queries
- At least 3 relational algebra expressions for selected queries

3. Functional Dependencies

Identify Functional Dependencies (FDs) for each table

4. Report

Prepare a professional report (in PDF or Word) that documents the full project. Follow the suggested structure:

Suggested Table of Contents

1. Introduction – Overview of the chosen topic and rationale for selecting this domain
2. Project Objectives – Define the project's goals and expected outcomes
3. Database Design – ER/EER Diagram, relational schema, and design justification
4. Query Execution – 15 diverse questions, SQL queries, explanations, and outputs

5. FDs
6. Screenshots, sample data, or additional notes

5. Deliverables

- Project Report (PDF or Word)
- Relational schema
- SQL Scripts: data.sql (sample data insertion), queries.sql (15 queries).
- ER/EER Diagram (draw.io, Lucidchart, or any modeling tool, exported as PNG or PDF)
- Relational Algebra document (for 3-5 queries)
- All deliverables should be compressed in one ZIP file and submitted (submission will be defined in a later announcement)

6. Group Size

Maximum of 2 students per group. Groups of more than 2 students will **NOT** be accepted

7. Submission and Presentation

- Submission Deadline: Last week of the semester
- Presentations will be held during the final week of the semester
- No hard copies are required